PROJECT – PART-I

Deliverable 2: Configuration Identification

Configuration Identification and Naming Convention Document

Web-based Invoice Trading Platform

Prepared using IEEE Std 828-2012 as the primary reference

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Document Control

Table 1: Document identification

Document field	Value
Project title	Web-based Invoice Trading Platform
Document title	Configuration Identification and Naming Convention Document
Identifier	ITP-CID-001 (CI-DOC-07)
Version / status	0.1 Draft submitted for instructor and team review; CCB approval pending
Plan owner	Configuration Manager
Governing plan	Software Configuration Management Plan, ITP-SCMP-001, version 0.1
Primary reference	IEEE Std 828-2012, Standard for Configuration Management in Systems and Software Engineering
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Document Objective

This document identifies the configuration items (CIs) of the Web-based Invoice Trading Platform, classifies them, assigns each a unique identifier, and defines the naming and version rules and the initial configuration baseline.

Revision History

Table 2: Revision history

Version	Change	Approval status
0.1	Initial configuration identification and naming convention document.	Pending CCB review

Approval Record

Table 3: CCB approval record

Approval field	Record
CCB meeting / date	Add after CCB decision
Participating members and affirmative votes	Add after CCB decision
Decision and conditions	Add after CCB decision
Approved document version / commit	Add after CCB decision

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1. Introduction

1.1 Purpose

This document records how the CIs of the Web-based Invoice Trading Platform are identified, classified, named, versioned and baselined. It is written for the team members who produce and control CIs, the Change Control Board (CCB) and the instructor who reviews the project.

1.2 Scope and Relationship to the SCMP

The document covers all project CIs from requirements to final archive. It is the companion to the Software Configuration Management Plan (SCMP), ITP-SCMP-001, whose Section 4 sets the governing identification process. The inventory and rules here must agree with that plan; if they differ, the SCMP applies until a Change Request (CR) corrects the difference. Abbreviations are defined in SCMP Section 1.4.

1.3 Current Project State

At this issue 21 of the 28 CIs exist as version 0.1 drafts awaiting team review. The other 7 are planned: source code, database scripts and automated tests have reserved file names only (Section 5.3), and no test report, release or status report exists yet. A planned entry reserves an identifier and owner; it does not mean the artifact exists.

1.4 Document Map

Table 4: Identification activities and deliverables

Activity or deliverable	Where addressed
Identify the major configuration items	Section 2
Classify configuration items into categories	Section 3
Assign a unique identifier to each configuration item	Section 4
Configuration Item (CI) Inventory	Section 5; CI register in the repository
Naming conventions and version identification rules	Sections 6 and 7
Initial configuration baseline	Section 8
GitHub repository containing the identified artifacts	Section 9

1.5 References

- **[R1]** IEEE Std 828-2012, Standard for Configuration Management in Systems and Software Engineering, Clause 8 (configuration identification).
- **[R2]** ITP-SCMP-001, Software Configuration Management Plan, Web-based Invoice Trading Platform, version 0.1.

2. Identification of Major Configuration Items

2.1 Selection Criteria

An item is a CI when it defines agreed scope or interfaces; is needed to build, test, deploy or recover the product; requires independent approval or ownership; is delivered to the instructor or users; or provides traceability and configuration evidence. Team members select CIs with the producers of the artifacts. An item is split into separate CIs when ownership, approval, independent release or risk requires it.

2.2 Major Items and Product Structure

The structure below follows the architecture of a React frontend, a Node.js / Express REST API and a PostgreSQL database. The product contains no firmware or custom hardware.

Table 5: Major configuration items and structure

Group	Major items	CIs	Relationship
Product	Each baseline and release of the platform	BL-00 to BL-5	Composite of the items below.
Application	Frontend, backend API, database schema and migrations	CI-SRC-01 to CI-SRC-03	Frontend calls the backend through CI-DOC-12; the backend uses the schema.
Verification	Automated tests, synthetic data, test plan, reports, traceability	CI-SRC-04, CI-DAT-01, CI-DOC-05, CI-DOC-08, CI-DOC-11	Show that requirements are implemented.
Definition	Charter, SRS, SDD, UI design, database design, interface contracts	CI-DOC-01 to CI-DOC-03, CI-DOC-09, CI-DOC-10, CI-DOC-12	Define what is built.
Build and environment	Workflows, environment templates, lock files, scripts, governance, tool versions	CI-CFG-01 to CI-CFG-06	Produce builds and record the tool chain.
Delivery	User manual, release notes and manifests	CI-DOC-06, CI-REL-01	Support delivery and use.
SCM control	SCMP, this document, CRs, registers, reports	CI-DOC-04, CI-DOC-07, CI-REC-01 to CI-REC-04	Record decisions and evidence.
Third-party items	Libraries, actions and hosted services	CI-SUP items in CI-REC-03	Accepted under SCMP Section 9.

2.3 Description of Each CI

The producer of a CI supplies its description and maintenance information. The Configuration Manager records them in the CI register (CI-REC-02) with the fields in SCMP Section 4.2. Source files are grouped by component and identified exactly through their Git path and commit.

3. Classification of Configuration Items

Every CI belongs to one category. Categories 1 to 7 are the required classes; the last two hold project control items that do not fit them.

Table 6: CI categories

No.	Category	CIs	Count
1	Requirements documents	CI-DOC-02	1
2	Design documents and diagrams	CI-DOC-03, CI-DOC-09, CI-DOC-10, CI-DOC-12	4
3	Source code	CI-SRC-01, CI-SRC-02	2
4	Database schemas and scripts	CI-SRC-03	1
5	Test cases and test reports	CI-DOC-05, CI-DOC-08, CI-DOC-11, CI-SRC-04, CI-DAT-01	5
6	Configuration files	CI-CFG-01, CI-CFG-02, CI-CFG-03, CI-CFG-04, CI-CFG-05, CI-CFG-06	6
7	User documentation	CI-DOC-06	1
8	Project and SCM documents	CI-DOC-01, CI-DOC-04, CI-DOC-07	3
9	Release and SCM records	CI-REL-01, CI-REC-01, CI-REC-02, CI-REC-03, CI-REC-04	5
	Total		28

4. Unique Identifiers

Each CI has one permanent identifier in the form **CI-<prefix>-<number>**, for example CI-DOC-02.

Table 7: Identifier prefixes

Prefix	Used for	Number format
DOC	Project, requirements, design, test and user documents	Two digits: CI-DOC-01
SRC	Source code, database scripts and automated tests	Two digits: CI-SRC-01
DAT	Synthetic test data	Two digits: CI-DAT-01
CFG	Workflows, environment templates, lock files, scripts and governance	Two digits: CI-CFG-01
REL	Release notes and delivery manifests	Two digits: CI-REL-01
REC	Change, register and report records	Two digits: CI-REC-01
SUP	Third-party supplier items	Three digits: CI-SUP-001

- The Configuration Manager allocates each identifier and records it in the CI register.
- Numbers are issued in sequence within a prefix. An identifier is never changed or reused, including after a CI is retired.
- New versions keep the same identifier; only the version changes.

5. Configuration Item Inventory

5.1 Inventory

Status values are Planned, Draft, Under review, Approved, Baselined, Released and Retired. "First baseline" is the earliest baseline planned to include the CI.

Table 8: Configuration item inventory

CI ID	Item	Controlled location	Owner	Version	Status	First baseline
CI-DOC-01	Charter	docs/project/	Project Manager	0.1	Draft	BL-00
CI-DOC-02	SRS	docs/requirements/	QA	0.1	Draft	BL-1
CI-DOC-03	SDD	docs/design/	Leads	0.1	Draft	BL-2
CI-DOC-04	SCMP	docs/scm/	Project Manager	0.1	Draft	BL-00
CI-DOC-05	Test plan/manual cases	docs/testing/	QA	0.1	Draft	BL-3
CI-DOC-06	User manual	docs/user/	QA	0.1	Draft	BL-5
CI-DOC-07	Configuration Identification Document	docs/scm/	Configuration Manager	0.1	Draft	BL-00
CI-DOC-08	Traceability matrix	docs/testing/	QA	0.1	Draft	BL-2
CI-DOC-09	UI design	docs/design/ui/	Frontend Lead	0.1	Draft	BL-2
CI-DOC-10	Database design/dictionary	docs/design/database/	Backend Lead	0.1	Draft	BL-2
CI-DOC-11	Test reports	docs/testing/reports/	QA	-	Planned	BL-3
CI-DOC-12	Interface contracts	docs/design/interfaces/	Both leads	0.1	Draft	BL-2
CI-SRC-01	Frontend	frontend/	Frontend Lead	-	Planned	BL-3
CI-SRC-02	Backend	backend/	Backend Lead	-	Planned	BL-3
CI-SRC-03	Schema/migrations/seeds	database/	Backend Lead	-	Planned	BL-3
CI-SRC-04	Automated tests	tests/	QA	-	Planned	BL-3
CI-DAT-01	Synthetic data	tests/data/	QA	0.1	Draft	BL-3
CI-CFG-01	CI workflows	.github/workflows/	Build Engineer	0.1	Draft	BL-00
CI-CFG-02	Environment templates	frontend/.env.example; backend/.env.example	Build Engineer	0.1	Draft	BL-00
CI-CFG-03	Dependency manifests/lock files	frontend/; backend/	Leads	0.1	Draft	BL-00
CI-CFG-04	Build/deploy scripts	scripts/	Build Engineer	0.1	Draft	BL-00
CI-CFG-05	Repository governance	.github/; .gitignore; README.md	Configuration Manager	0.1	Draft	BL-00
CI-CFG-06	Environment/tool version record	docs/scm/environments/	Build Engineer	0.1	Draft	BL-00
CI-REL-01	Release notes/delivery manifests	release/	Configuration Manager	-	Planned	BL-5
CI-REC-01	CRs/decisions	GitHub Issues; docs/scm/ccb/	Configuration Manager	0.1	Draft	BL-00

CI ID	Item	Controlled location	Owner	Version	Status	First baseline
CI-REC-02	CI register/baselines	docs/scm/registers/; docs/scm/baselines/	Configuration Manager	0.1	Draft	BL-00
CI-REC-03	Supplier/dependency acceptance register	docs/scm/suppliers/	Leads	0.1	Draft	BL-00
CI-REC-04	Status, surveillance and audit reports	docs/scm/reports/	Configuration Manager	-	Planned	BL-1

5.2 Status Summary

Table 9: Inventory status at this issue

Status	Count	CIs
Draft	21	CI-DOC-01, CI-DOC-02, CI-DOC-03, CI-DOC-04, CI-DOC-05, CI-DOC-06, CI-DOC-07, CI-DOC-08, CI-DOC-09, CI-DOC-10, CI-DOC-12, CI-DAT-01, CI-CFG-01, CI-CFG-02, CI-CFG-03, CI-CFG-04, CI-CFG-05, CI-CFG-06, CI-REC-01, CI-REC-02, CI-REC-03
Planned	7	CI-DOC-11, CI-SRC-01, CI-SRC-02, CI-SRC-03, CI-SRC-04, CI-REL-01, CI-REC-04

5.3 CI Register and Reserved File Names

The complete register, with description, backup owner, origin, Git reference, parent CIs and other SCMP fields, is held at docs/scm/registers/ITP-REG-CIRegister-v0.1.csv. Third-party items are held in docs/scm/suppliers/ITP-REG-Suppliers-v0.1.csv.

These files exist in the repository under their final names but are empty until their content is added.

Table 10: Reserved source file names

CI ID	Files
CI-SRC-01	frontend/index.html; frontend/src/App.jsx; frontend/src/components/AdminDashboard.jsx; frontend/src/components/AuthPanel.jsx; frontend/src/components/InvestorDashboard.jsx; frontend/src/components/InvoiceUploadForm.jsx; frontend/src/components/SellerDashboard.jsx; frontend/src/components/SettlementList.jsx; frontend/src/main.jsx; frontend/src/services/api-client.js; frontend/src/styles.css; frontend/vite.config.js
CI-SRC-02	backend/jest.config.js; backend/src/app.js; backend/src/config.js; backend/src/db.js; backend/src/middleware/auth-middleware.js; backend/src/middleware/error-middleware.js; backend/src/migrate.js; backend/src/routes/auth-routes.js; backend/src/routes/bid-routes.js; backend/src/routes/invoice-routes.js; backend/src/routes/report-routes.js; backend/src/routes/settlement-routes.js; backend/src/seed.js; backend/src/server.js; backend/src/utils/validation.js
CI-SRC-03	database/migrations/001_create_users.sql; database/migrations/002_create_invoices.sql; database/migrations/003_create_bids.sql; database/migrations/004_create_settlements.sql; database/seeds/001_demo_data.sql
CI-SRC-04	tests/auth-routes.test.js; tests/bid-routes.test.js; tests/helpers/setup-env.js; tests/helpers/test-app.js; tests/invoice-routes.test.js; tests/report-routes.test.js; tests/settlement-routes.test.js

6. Naming Conventions

6.1 General Rules

Table 11: Naming rules

Item	Rule	Example
Document files	ITP-<Type>-<Title>-v<Major>.<Minor>.<extension>	ITP-SRS-Requirements-v1.0.docx
Folders and general JS	Lowercase, hyphenated	invoice-management/; invoice-service.js
React components	PascalCase	InvoiceUploadForm.jsx
Functions / constants	camelCase / UPPER_SNAKE_CASE	calculateDiscount; MAX_BID_COUNT
Tests	<unit>.test.js	invoice-service.test.js
Database migrations	<sequence>_<action>_<object>.sql	001_create_users.sql
Branches	<type>/<short-name>; types feature, bugfix, hotfix, docs, release	feature/invoice-upload; release/1.0.0
Commits	<type>(<scope>): <description>; types feat, fix, docs, test, refactor, style, chore	feat(invoice): add upload form
Requirements and tests	FR-nn; NFR-nn; TC-nn	FR-01; NFR-01; TC-01
Change Requests	CR-nnn, mapped to the GitHub Issue	CR-001
Interfaces	IF-nn	IF-01
Baselines	ITP-BL-<nn>-<name> as the Git tag	ITP-BL-00-initial
Builds	<version>+<workflow run>.<short commit>	1.0.0-rc.1+42.a1b2c3d
Third-party items	CI-SUP-nnn; supplier name and exact version retained	CI-SUP-001 actions/checkout v4.3.1

6.2 Document Type Codes

Table 12: Document type codes

Code	Document	CI	File name
CHR	Project charter	CI-DOC-01	ITP-CHR-Charter-v0.1.md
SRS	Software Requirements Specification	CI-DOC-02	ITP-SRS-Requirements-v0.1.md
SDD	Software Design Document	CI-DOC-03	ITP-SDD-Design-v0.1.md
SCMP	Configuration Management Plan	CI-DOC-04	ITP-SCMP-Plan-v0.1.docx
TST	Test plan and manual cases	CI-DOC-05	ITP-TST-TestPlan-v0.1.md
USR	User manual	CI-DOC-06	ITP-USR-UserManual-v0.1.md
CID	Configuration identification document	CI-DOC-07	ITP-CID-Identification-v0.1.docx
TRM	Traceability matrix	CI-DOC-08	ITP-TRM-TraceabilityMatrix-v0.1.csv
UID	User interface design	CI-DOC-09	ITP-UID-InterfaceDesign-v0.1.md
DBD	Database design and dictionary	CI-DOC-10	ITP-DBD-DatabaseDesign-v0.1.md
RPT	Test or SCM report	CI-DOC-11; CI-REC-04	ITP-RPT-<Title>-v0.1.md

Code	Document	CI	File name
IFC	Interface contract	CI-DOC-12	ITP-IFC-ApiContract-v0.1.md
ENV	Environment and tool version record	CI-CFG-06	ITP-ENV-ToolVersions-v0.1.md
REG	Register	CI-REC-02; CI-REC-03	ITP-REG-CIRegister-v0.1.csv
MAN	Baseline or release manifest	CI-REC-02; CI-REL-01	ITP-MAN-BL00-v0.1.md

7. Version Identification

Table 13: Version identification rules

Item	Rule
Software	MAJOR.MINOR.PATCH. MAJOR for incompatible interface or behavior changes, MINOR for backward-compatible functions, PATCH for backward-compatible fixes.
Pre-release software	Suffix -rc.n, for example 1.0.0-rc.1.
Documents	0.x draft; 1.0 first approval; 1.1 minor approved update; 2.0 substantial approved revision.
Source files	CI grouping, Git path, commit and applicable baseline or release tag. No separate version is kept for each file.
Database	Migration sequence number. A baselined migration that has been applied is corrected only by a new migration.
Records and registers	Document version in the file name; Git history keeps each change.
Third-party items	Exact version and immutable reference where available, such as a commit SHA or lock-file entry.

- An approved baseline or release tag is never moved or reused. A new approved configuration receives a new identifier.
- The editable document is authoritative. An exported PDF must match its approved revision.

8. Initial Configuration Baseline

8.1 Definition of BL-00

BL-00 is the first formal reference point of the project, and it fixes the agreed plans, inventory, rules and repository setup at start-up. It contains only artifacts that exist and have been reviewed. Later work and changes are measured against it.

8.2 Contents

Table 14: BL-00 contents

CI ID	Repository path	Version
CI-DOC-01	docs/project/ITP-CHR-Charter-v0.1.md	0.1
CI-DOC-04	docs/scm/ITP-SCMP-Plan-v0.1.docx	0.1
CI-DOC-07	docs/scm/ITP-CID-Identification-v0.1.docx	0.1
CI-CFG-03	frontend/package.json; frontend/package-lock.json; backend/package.json; backend/package-lock.json	0.1

CI ID	Repository path	Version
CI-CFG-01	.github/workflows/ci.yml	0.1
CI-CFG-02	frontend/.env.example; backend/.env.example	0.1
CI-CFG-04	scripts/check-structure.sh	0.1
CI-CFG-05	README.md; .gitignore; .github/CONTRIBUTING.md; .github/PULL_REQUEST_TEMPLATE.md; .github/ISSUE_TEMPLATE/	0.1
CI-CFG-06	docs/scm/environments/ITP-ENV-ToolVersions-v0.1.md	0.1
CI-REC-01	docs/scm/ccb/CCB-001-BL-00.md	0.1
CI-REC-02	docs/scm/registers/ITP-REG-CIRegister-v0.1.csv; docs/scm/baselines/ITP-MAN-BL00-v0.1.md	0.1
CI-REC-03	docs/scm/suppliers/ITP-REG-Suppliers-v0.1.csv	0.1

BL-00 includes the existing plans, charter, registers and configuration. The SRS, design documents, test plan, traceability matrix and user manual exist as drafts in the repository but are excluded because they are approved at the later baselines in Section 8.5. Source code, database scripts and automated tests are excluded because only their reserved file names exist.

8.3 Establishing the Baseline

Table 15: Steps to establish BL-00

Step	Action	Responsible
1	Upload the BL-00 files to the repository. This first import to main is the only direct commit and is logged as a deviation in the manifest; PR control applies afterwards.	Configuration Manager
2	Add the repository URL to README.md, this document and the manifest.	Configuration Manager
3	Review each BL-00 item for completeness and agreement with the SCMP.	QA; a non-author reviewer checks items written by QA
4	Perform the baseline configuration audit: only ready items included, identifiers correct, approvals recorded.	Non-author reviewer
5	Hold the CCB review and record the decision in docs/scm/ccb/CCB-001-BL-00.md. Approval needs at least three affirmative votes, including the Project Manager or Configuration Manager.	CCB
6	Complete the manifest with the exact commit, then apply the tag ITP-BL-00-initial.	Configuration Manager
7	Create a Git bundle in separate storage, verify it and record the backup in the manifest.	Configuration Manager

8.4 Baseline Record

Table 16: BL-00 record

Field	Record
Baseline identifier	BL-00
Git tag	ITP-BL-00-initial
Status	Prepared; CCB approval pending. The tag is applied only after approval.
Repository URL	https://github.com/arsema16/invoice-trading-platform.git

Field	Record
Approved commit	Add after CCB decision
CCB decision reference and date	Add after CCB decision
Manifest	docs/scm/baselines/ITP-MAN-BL00-v0.1.md

8.5 Later Baselines

Table 17: Planned baselines

Baseline	Git tag	Trigger and new CIs
BL-1	ITP-BL-01-requirements	SRS approved: CI-DOC-02, CI-REC-04.
BL-2	ITP-BL-02-design	Design approved: CI-DOC-03, CI-DOC-08, CI-DOC-09, CI-DOC-10, CI-DOC-12.
BL-3	ITP-BL-03-development	Scoped code and unit/API tests pass: CI-DOC-05, CI-DOC-11, CI-SRC-01, CI-SRC-02, CI-SRC-03, CI-SRC-04, CI-DAT-01.
BL-4	ITP-BL-04-testing	System and acceptance evidence complete and FCA findings resolved; updates to existing CIs.
BL-5	ITP-BL-05-release	PCA passed and release approved: CI-DOC-06, CI-REL-01.

9. GitHub Repository

The repository is named invoice-trading-platform. Its URL is https://github.com/arsema16/invoice-trading-platform.git. It uses the protected branches main and develop, with work done on the branch types in Section 6.1.

Table 18: Repository structure

Location	Contents	CIs
README.md; .gitignore; .github/	Project overview, ignore rules, contribution rules, issue and PR templates	CI-CFG-05
.github/workflows/	Automated repository checks	CI-CFG-01
docs/project/	Charter	CI-DOC-01
docs/requirements/	SRS	CI-DOC-02
docs/design/	SDD and diagrams; ui/, database/ and interfaces/ subfolders	CI-DOC-03, 09, 10, 12
docs/testing/	Test plan, traceability matrix; reports/ subfolder	CI-DOC-05, 08, 11
docs/user/	User manual	CI-DOC-06
docs/scm/	SCMP, this document; registers/, baselines/, ccb/, suppliers/, environments/, reports/ subfolders	CI-DOC-04, 07; CI-REC-01 to 04; CI-CFG-06
frontend/; backend/	Application source, environment templates and lock files	CI-SRC-01, 02; CI-CFG-02, 03
database/	Schema, migrations and seeds	CI-SRC-03
tests/; tests/data/	Automated tests and synthetic data	CI-SRC-04; CI-DAT-01
scripts/	Build, deployment and check scripts	CI-CFG-04
release/	Release notes and delivery manifests	CI-REL-01

No secrets or real financial data are stored in the repository. Environment templates contain placeholders only.

10. Maintenance of this Document

- The Configuration Manager owns this document and updates it when a CI is added, split or retired, or when a rule changes.
- A new CI follows the acquisition steps in SCMP Section 4.2 before it is added to the inventory.
- After BL-00, changes to this document require a CR and CCB approval.
- QA checks that this document, the CI register and the SCMP agree at each phase boundary.